

KUO-HAN HUNG

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Education

Stanford University

M.S. in Computer Science

June 2027 (Expected)

Stanford, CA

- GPA: 4.0/4.0; Specialization: Artificial Intelligence and Robotics.

National Taiwan University

B.S. in Computer Science and Information Engineering

Dec. 2024

Taipei, Taiwan

- GPA: 4.24/4.3, Top 3% of Class. **Additional Special Program:** Creativity and Entrepreneurship.

Technical Skills

- **Robotics:** Real-Robot Deployment and Training, Real-to-Sim, Real-Time Control, Manipulation, ROS.
- **Parallel Computing:** Distributed Training, Ray, Multiprocessing/Asyncio, Vectorized Environments.
- **Languages & Frameworks:** Python, C/C++, Golang, JavaScript, SQL, Bash, PyTorch, JAX, NumPy.
- **Infrastructure:** Linux, Docker, SLURM, Git, GCP, Azure.

Research Experience

Stanford Artificial Intelligence Laboratory (ILIAD & IRIS Labs)

Sept. 2025 – Present

Graduate Researcher, Advisors: Prof. Dorsa Sadigh & Prof. Chelsea Finn

Stanford, CA

- **RL finetuning of real-time VLA policies** under inference latency. [1]
 - Decoupled slow VLA proposals from a fast edit policy reacting to the latest observation, raising success from 42% to 97% across six tasks including dexterous light-string routing and flower insertion, with under 10 min of online data.
 - Built the full real-time policy stack enabling effective RL inference and training: background-thread VLA inference overlapped with chunk execution, action-prefix inpainting, and a 30 Hz Cartesian control loop.
- **Sample-efficient RL finetuning of pretrained VLAs** in the real world. [2]
 - Stabilized RL finetuning with residual action edits and Q-guided sampling, reaching 30/30 success across six real-world tasks including dexterous light-string routing and flower insertion, with 19 min of online data.
 - Designed and built the real-world RL pipeline: asynchronous multi-GPU training, WebSocket server-client rollout, replay buffer, data pipelines, and DROID integration (100+ GitHub stars).
- **Scaling Q-functions** with Transformers in offline RL. [3]
 - Introduced attention entropy regularization to stabilize and scale Transformer value functions.
 - Showed that all prior methods fail to scale with increased model capacity, while TQL gains 43%.
- **Visual representation pretraining** for reinforcement learning (ongoing).

Embodied AI Lab

June 2025 – Sept. 2025

Undergraduate Researcher, Advisor: Prof. Tsung-Wei Ke

Taipei, Taiwan

- **Generalized dexterous manipulation policies** via digital twins. [6]
 - Built a real-to-sim pipeline: videos to digital twins via mesh generation and object/human pose estimation.
 - Trained residual RL policies on digital twins from in-the-wild videos, yielding diverse dexterous behaviors.

Robot Learning Lab

Oct. 2024 – May 2025

Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun

Taipei, Taiwan

- **Manipulation policies learned from action-free videos.** [5]
 - Designed a Transformer to learn skills from action-free videos, using optical flow as an action surrogate.
 - Enabled multi-task and long-horizon planning by translating flow-based skill plans into executable actions.

Communication and Multimedia Lab

May 2022 – May 2024

Undergraduate Researcher, Advisors: Prof. Winston H. Hsu & Prof. Yi-Ting Chen

Taipei, Taiwan

- **Vision-instruction correlation rewards** for long-horizon manipulation. [4]
 - Finetuned LLMs and VLMs into a hierarchical video reward model for generating stage-aware signals.
 - Achieved a 43% improvement in success rate over state-of-the-art baselines on long-horizon tasks.
- **Behavior error detection** in few-shot imitation policies via representation pattern detection. [8]

Selected Publications

(* indicates equal contribution)

- [1] **Reinforcement Learning for Real-Time Vision-Language-Action Policies**
Perry Dong*, **Kuo-Han Hung***, Dorsa Sadigh, and Chelsea Finn
Under Review
- [2] **EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models**
Perry Dong*, **Kuo-Han Hung***, Tian Gao, Dorsa Sadigh, and Chelsea Finn
Conference on Robot Learning (CoRL), 2026
- [3] **TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse**
Perry Dong*, **Kuo-Han Hung***, Alexander Swerdlow, Dorsa Sadigh, and Chelsea Finn
International Conference on Machine Learning (ICML), 2026
- [4] **VICtoR: Learning Hierarchical Vision-Instruction Rewards for Long-horizon Manipulation**
Kuo-Han Hung*, Pang-Chi Lo*, Jia-Fong Yeh*, Han-Yuan Hsu, Yi-Ting Chen, and Winston Hsu
International Conference on Learning Representations (ICLR), 2025
- [5] **Learning Skill Abstraction from Action-Free Videos via Optical Flow**
Hung-Chieh Fang*, **Kuo-Han Hung***, Chu-Rong Chen, ..., Min-Hung Chen, and Shao-Hua Sun
ICML Workshop on Building Physically Plausible World Models, 2025
- [6] **DexMan: Learning Bimanual Dexterous Manipulation from Human and Generated Videos**
Jhen Hsieh, Kuan-Hsun Tu, **Kuo-Han Hung**, and Tsung-Wei Ke
Under Review
- [7] **Attention Tracker: Detecting Prompt Injection Attacks in LLMs**
Kuo-Han Hung, Ching-Yun Ko, Ambrish Rawat, I-Hsin Chung, Winston Hsu, and Pin-Yu Chen
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), 2025
- [8] **AED: Adaptable Error Detection for Few-shot Imitation Policy**
Jia-Fong Yeh, **Kuo-Han Hung**, Pang-Chi Lo, Chi-Ming Chung, ..., Yi-Ting Chen, and Winston Hsu
Conference on Neural Information Processing Systems (NeurIPS), 2024
- [9] **Customizable Routing with Learning from Past Recommendations**
Kuo-Han Hung, Chiqun Zhang, and Dragomir Yankov
International Conference on Advances in Geographic Information Systems (SIGSPATIAL), 2024

Work Experience

IBM Research

June 2024 – Sept. 2024

Research Intern, Host: Dr. Pin-Yu Chen

Yorktown Heights, NY

- Found that attention drifts from the original instruction during **prompt injection attacks on LLM agents**. [7]
- Traced the effect to specific attention heads using mechanistic interpretability.
- Designed an attention-based detector, a 31% AUROC improvement over the prior state of the art.

Microsoft

July 2023 – June 2024

Applied Scientist Intern, Host: Dr. Chiqun Zhang

Mountain View, CA (Remote)

- **Integrated Bing Maps with Microsoft Copilot** for conversational, multi-destination map queries.
- Formulated a pathfinding algorithm that learns from history, **achieving a 10× speedup over Dijkstra**. [9]
- Developed an LLM-powered search system that adapts to user intent and geographic context.

Awards & Honors

- **Student Research Scholarship**, Taiwan National Science Council. Sept. 2024
- **Bachelor Thesis Dean's Award**, National Taiwan University — top 2 theses in the CS department. May 2024
- **Lin Hsiung Chen Memorial Scholarship** — for top 1 CS student in NTU. Nov. 2023
- **Creativity and Entrepreneurship Excellence Award**, NTU D-School. June 2023
- **Presidential Award** (×2), National Taiwan University — top 2% of the CS department. Sept. 2022, 2023
- **Dean's List** (×3), National Taiwan University — top 5% of the CS department. June 2021, 2022, 2023